TRAINER KEY. DO NOT DISTRIBUTE. Day 3, Session 2: Same numbers. Different story. (charts and captions)

Source: H8-A. Figure C: mean cost of the same 4,794 participants 1,432 before and 763 after (fall 669). Chart A axis starts at zero; Chart B at 700 AED; same numbers, different impression. Neither chart shows a causal effect or that every business fell. Distribution spread (Figure D: middle 50% 1,096 to 1,694) is variation between businesses, not a confidence interval. Percentages (Q4): small business 10 to 2 tonnes (80%), large 90 to 54 (40%); total 100 to 56 (44%); mean of the two percentages 60%; weighted by starting tonnes 44%. City landfill: 49.5% (fall in total tonnes; 31,766.4 to 16,038.0) and 58.6% (mean of businesses' percentage falls). Pilot (Figure P): reduction 1,014, interval 862 to 1,167, which includes values below 1,000.

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H8-A §1: Reading the bar charts

GreenWaste is a fictional training case. Use H8-A (the bar charts and distributions) as your source for today's questions.

Figure C: Mean annual waste management cost of the same participant businesses, drawn twice. Chart A starts at zero. Chart B starts at 700 AED.

Chart A: the vertical axis starts at zero



Chart B: the vertical axis starts at 700 AED



Table C: The values behind Figure C: mean annual waste management cost of the same 4,794 city participant businesses (AED)

Group	Before GreenWaste (AED)	12 months after GreenWaste (AED)	Fall in mean cost (AED)
Same 4,794 city participant businesses	1,432	763	669

These are the mean annual waste management costs, in AED per business, of the same 4,794 city participant businesses before GreenWaste and 12 months after. The bars compare two times for participants. They do not compare participants with comparison businesses that did not receive GreenWaste. The values are rounded. The mean fall is about **669 AED**.

What is supported, and what is not?

The bars describe the participants' mean cost before and after GreenWaste. They do not show that GreenWaste caused the fall. They do not show that every participant's cost fell, or that the mean describes every business. The bars do not show individual changes, a credible comparison for a causal reading, or the uncertainty of an estimated effect.

Why does the baseline matter?

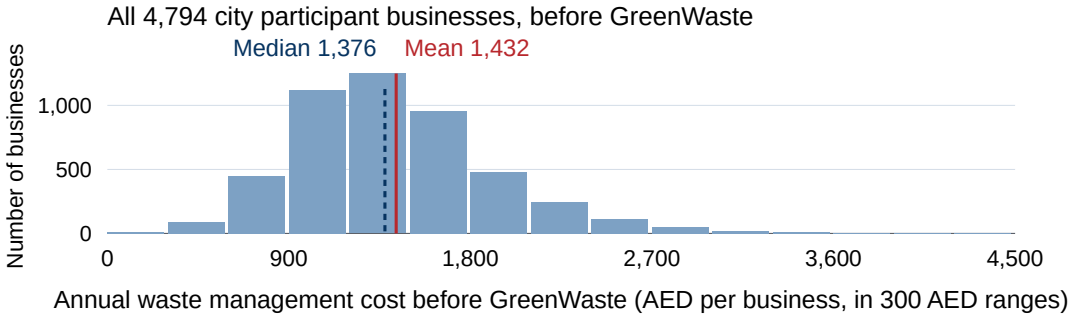
The length of a bar shows an amount. A baseline at zero keeps the lengths in proportion to the amounts. Chart B starts at 700 AED, so its bar lengths show the amounts above 700 AED, not the full costs. The labels give the values, but they may not change a reader’s first impression. A baseline that does not start at zero can be appropriate for other types of chart, or when it is clearly explained. It is not automatically misleading.

The sentence and the audience also matter. A large drop in the height of the bars does not show an effect of the programme. A better caption cannot create evidence that is missing.

Sources: Figure C and Table C in H8-A §1. Both charts have the same upper bound, 1,600 AED. Only the lower bound differs.

Page 2 | H8-A §2: Distributions show variation between businesses, not uncertainty

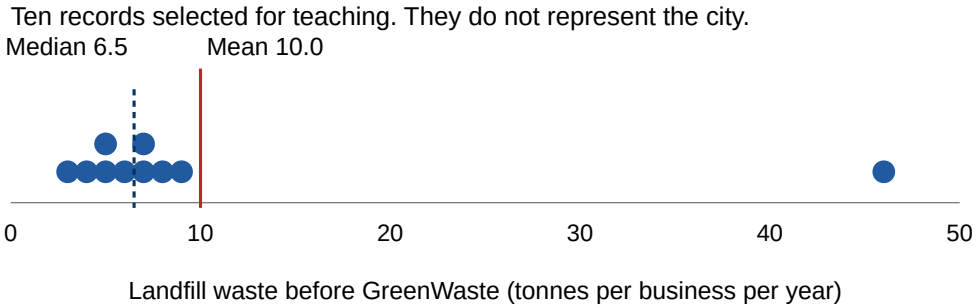
Figure D: Annual waste management cost before GreenWaste of all 4,794 city participant businesses (AED)



The histogram includes all 4,794 city participant businesses. Each bar counts the businesses whose cost falls within a 300 AED range. The mean is **1,432 AED** and the median is **1,376 AED**. The middle 50% of businesses have costs from **1,096 to 1,694 AED**. The range is **153 to 4,306 AED**. These numbers describe how much costs varied between businesses before GreenWaste. They are not an estimated effect and not a confidence interval.

The outcome changes on this page. Figure E shows landfill waste in tonnes per business per year, not annual waste management cost in AED.

Figure E: Landfill waste before GreenWaste in ten selected records (tonnes per business per year)



The ten values are 3, 4, 5, 5, 6, 7, 7, 8, 9 and 46 tonnes per business per year. The mean is **10.0 tonnes**, the median is **6.5 tonnes** and the range is **3 to 46 tonnes**. Nine of the ten values are below the mean. Repeated values are shown as separate stacked dots. We selected these ten records for teaching. They do not represent the city. One large value can raise the mean, but that alone is not a reason to delete it.

Two different meanings of spread

A distribution shows how much businesses differ from each other. The interval for the pilot on page 7 shows how uncertain we are about an estimated mean effect. The middle 50% in Figure D and the range of the ten records in Figure E are not that interval.

Sources: H8-A (the bar charts and distributions) §2, Figures D and E. Figure D uses the before-GreenWaste cost of every participant business. Figure E uses ten records chosen earlier in the course. Page 7 changes the group of businesses and the outcome again, for the pilot.

Page 3 | Q1 to Q3: mark the chart and interpret it

Q1 | Mark Figure C on page 1

a. Circle the units and mark where each axis starts.

b. Underline the group of businesses, the time period and the comparison in the caption.

c. Use Table C to check the bars.

d. What changes when the baseline changes?

> The bar lengths and the visual size of the drop: with a baseline at 700 the after bar looks far shorter relative to the before bar, so the fall looks larger and the bars show amounts above 700, not the full costs.

e. What stays the same?

> The data and what they show: 1,432 before, 763 after, a fall of 669 AED for the same 4,794 participants; the axis labels; and what is not shown (no causal comparison, no individual changes).

Q2 | What do the bar charts show?

a. Write one descriptive statement that the bars support, and the source you took it from.

> Example: "The mean annual waste management cost of the same 4,794 city participant businesses was 1,432 AED before GreenWaste and 763 AED 12 months after, a fall of about 669 AED" (Table C / H8-A §1).

b. Write one specific limit, and the evidence you would need to address it.

> Example: it does not show that GreenWaste caused the fall; evidence needed: a credible comparison (for example comparison businesses or earlier trends) and its assumptions. Or: it does not show every participant fell; evidence: the distribution of individual changes.

Q3 | Read Figures D and E on page 2

a. What does the histogram of all participants add to the mean?

> It shows how much costs vary between businesses (spread and shape: from 153 to 4,306, median 1,376, middle 50% 1,096 to 1,694), which a mean alone hides.

b. Explain the middle 50% without calling it a confidence interval.

> Half of the businesses had costs between 1,096 and 1,694 AED before GreenWaste. It describes how businesses vary, not uncertainty about an estimated effect.

c. Why does the mean of the ten selected landfill records differ from their median?

> One very large value (46 tonnes) pulls the mean up to 10.0 tonnes, while the median stays at 6.5 tonnes; nine of the ten values are below the mean.

d. Why must we not call the ten records a result for the city?

> They are ten records selected for teaching, not a representative sample of the city, so the ten are not a result for the city.

e. What changes between Figure D and Figure E: the outcome, the units, the group of businesses, or more than one of these?

More than one: the outcome (landfill waste, not cost), the units (tonnes per business, not AED) and the group (ten selected records, not all 4,794 participants).

Page 4 | Q4 to Q7: percentages, the caption and your judgement

Q4 | Two invented businesses. Calculate first, then open page 7.

a. Complete the table: the fall in tonnes for each business as a percentage of its starting amount.

Business	Before (tonnes)	After (tonnes)	Fall (%)
Business with a small starting amount	10	2	**80%**
Business with a large starting amount	90	54	**40%**

b. Calculate the total tonnes before: **100 tonnes** and after: **56 tonnes**, and the fall in the total as a percentage: **44** %.

c. Calculate the mean of the two business percentage falls: **60** %.

d. Why are the two percentages different?

> The small business has a large percentage fall (80%) but few tonnes; the large business has a smaller percentage (40%) but most of the tonnes. The fall in total weights businesses by size; the mean of the two percentages weights each business equally.

e. After you see the summary for the city on page 7, explain what 49.5% and 58.6% each measure. Which total is each percentage based on?

> 49.5% = fall in total participant tonnes (31,766.4 to 16,038.0): larger starting amounts count for more. 58.6% = mean of each business's own percentage fall: each of the 4,794 counts equally; the base is each business's own starting amount.

Q5 | Repair the caption for the director

Draft caption: **“GreenWaste substantially reduced every participant’s costs.”**

Use Figure C. Write a caption that names the group of businesses, the outcome, the units, the time, the comparison, the source and the limit on causal claims. Choose a baseline that suits your sentence, and say why.

> Example: "Among 4,794 city participant businesses, mean annual waste management cost was about 1,432 AED before GreenWaste and 763 AED 12 months after (Source: H8-A, Table C). This is a description of two times and does not show the causal effect of the programme or a fall for every participant." Baseline: zero for a bar chart of full costs, so that bar lengths stay in proportion.

Q6 | Read the pilot interval (page 7, Figure P)

a. What uncertainty does the interval show?

> The uncertainty about the estimated mean reduction in the pilot (862 to 1,167 AED). It does not show the distribution of individual savings or every source of bias.

b. Does it show that our rule of at least a 1,000 AED reduction in annual waste management cost per business per year is met?

> No. The estimated reduction is 1,014 AED but the interval 862 to 1,167 includes values below 1,000, so it does not show that the rule is met.

Q7 | Close the source and give your first chart check

a. Name one check you will do before trusting a chart, and the mistake it prevents.

> Example: check the axis start and units, to avoid being misled by the visual size of a difference; or check the group and comparison named in the caption, to avoid causal claims from a before-and-after bar.

b. Write one statement that today’s evidence supports and one limit.

> Example: "Participants' mean cost was 1,432 before and 763 12 months after for the same 4,794 businesses; this does not show a cause or a fall for every business." Accept any supported statement and a limit.

Page 5 | AI exercise: chart, claim and audience

Stage 1: Mark the response

Naive prompt (a prompt with no context, as a busy person might type it): “Which of these two charts better represents the difference, and why?” The AI was given Figure C and the draft claim. This response was written for the exercise. It is not a real AI answer.

> Both charts show the same underlying participant means accurately: annual waste management cost was about 1,432 AED before and 763 AED after, a recorded fall of 669 AED.
>
> ~~The chart whose axis does not start at zero is often preferable for a policy audience because it makes the difference visible.~~ A zero baseline is not a requirement in professional visualisation, and the labels allow readers to recover the exact values.
>
> ~~On balance, B communicates the finding more effectively beside~~ "GreenWaste substantially reduced every participant's costs." ~~The large visual drop makes the programme effect clear; anyone needing the absolute values can consult the labels.~~

Keep the readings that H8-A (the bar charts and distributions) supports. Cross out the advice and conclusions that it does not support. Check whether each sentence answers the question about the claim and the audience.

> Supported: both charts show the same means accurately (1,432, 763, 669). Not supported: that the non-zero axis is preferable for a policy audience; that B suits the claim "substantially reduced every participant's costs"; that the large visual drop shows a programme effect; that labels fix the first impression. Part-supported: "a zero baseline is not a requirement" (true for some chart types, not a reason to choose B here).

Stage 2: Run the improved prompt in Microsoft Copilot

Improved prompt

These two bar charts of annual waste management cost will sit beside the claim "GreenWaste substantially reduced every participant's costs" in a report for a director who is not a technical expert. Using only H8-A below, ask me questions before you recommend a chart. Check whether the before and after means support that sentence, whether the words "every" and "programme effect" are justified, and how a zero baseline compared with a baseline of 700 AED changes how large the difference looks. Separate whether a chart is numerically accurate from whether it suits this claim and audience. Do not say that every axis that does not start at zero is wrong. Do not confuse the spread of a distribution with a confidence interval. Suggest a corrected caption that names the group of businesses, the outcome, the units, the period, the comparison and the limit on causal claims. Cite the source for each point. Source: [paste H8-A pages 1 and 2; attach or show the two charts of Figure C from page 1 if you can].

Use only the information in H8-A, and keep your AI good-practice card beside you. If you are not using Copilot, read the prepared response on page 6 after you finish Stage 1.

a. Record one correction you verified.

> Example: the means (1,432 and 763) are right but "every participant" and "programme effect" are not supported; or B's axis exaggerates the drop. Accept any correction verified against H8-A.

b. Record one useful question the response asked about missing evidence.

> Example: "What decision will the director make from this figure?" or "Is there causal evidence elsewhere?"

Page 6 | Prepared response

Use this response if you are not using Copilot. Read it after you have marked the response on page 5. It was written for the exercise. It is not a real AI answer. An AI tool may answer differently.

Questions before choosing the display

What decision should the director make from this figure? Is there credible causal evidence elsewhere, or must this sentence remain a description? Is the claim about an average, about every participant, or about value for money for the whole programme?

Check the numbers and what the chart claims

H8-A §1, Table C and Figure C show the same mean annual waste management costs before and after, about **1,432 and 763 AED**, for the same **4,794 city participant businesses**. The mean fall is about **669 AED**.

Chart B's lower bound of 700 AED makes the after bar look small compared with the before bar. Its labels are numerically correct, but that does not make the dramatic shape appropriate beside a claim about causes or about every participant, for a reader who is not an expert.

For a bar chart of these full costs, Chart A's zero baseline is a safer choice. Other chart types may use a clearly explained axis that does not start at zero, so it is not necessary to ban every such axis.

Correct the sentence, not just the chart

"Among 4,794 city participant businesses, mean annual waste management cost was about 1,432 AED before GreenWaste and 763 AED 12 months after. Source: H8-A §1. This description of two times does not show the causal effect of the programme, and it does not show a reduction for every participant."

H8-A §2 shows variation in costs before GreenWaste. It does not show individual changes, or the uncertainty about an effect of the programme. To assess a decision about expansion, request the relevant causal comparison and its assumptions, the target population and the economic evidence. A clearer chart cannot provide those missing facts.

Verify this response yourself

a. Check one factual reading against its source.

> Accept any reading checked against Table C, Figure C or §§2 to 4 (e.g. 1,432 to 763, fall 669; Chart B starts at 700 AED).

b. Write one reason the chart choice fits the corrected claim, and one request for evidence.

> Zero baseline keeps bar lengths in proportion for a description of full costs. Request: a credible causal comparison and its assumptions, or the target population or economic evidence.

Page 7 | H8-A §§3 and 4: numbers for reference

Open this page after you have completed the two-business calculation in Q4.

§3 | Percentage falls: an invented example and the summary for the city

Business	Before (tonnes)	After (tonnes)	Fall (%)
Business with a small starting amount	10	2	80
Business with a large starting amount	90	54	40

In the invented example, total tonnes fall from 100 to 56, which is a fall of **44%**. The mean of the two business percentage falls is (80% + 40%) / 2 = **60%**. Weighting each business by its starting tonnes gives 0.1 × 80% + 0.9 × 40% = **44%**. Every starting amount in this example is above zero, so each percentage can be calculated.

Landfill waste of the same 4,794 city participant businesses, before GreenWaste and 12 months after: two ways to summarise the fall

Summary	Value	How businesses are weighted
Fall in total participant tonnes	49.5%	Businesses with larger starting amounts count for more
Mean of the percentage falls of the participant businesses	58.6%	Each of the 4,794 businesses counts equally

The total before GreenWaste was **31,766.4 tonnes** and the total after was **16,038.0 tonnes**. Every business had a landfill amount above zero before GreenWaste. Neither summary alone shows that GreenWaste caused the fall, or that every business fell. One summary gives more weight to businesses with larger starting amounts. The other gives every business the same weight.

§4 | Figure P: the separate pilot and annual waste management cost

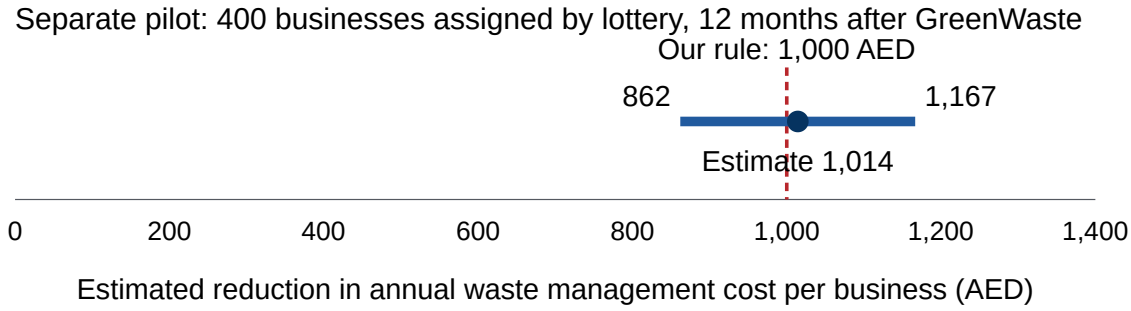


Figure P: Estimated mean reduction in annual waste management cost in the pilot, with its 95% interval, compared with our rule (AED)

This is the separate pilot of 400 businesses. A lottery assigned 200 businesses to the treatment group and 200 to the control group. The comparison is of annual waste management cost 12 months after GreenWaste. The estimated mean reduction is **1,014 AED**, with a 95% interval of **862 to 1,167 AED**. The interval includes values below our rule of at least a 1,000 AED reduction. The interval shows uncertainty about the estimated mean effect. It does not show the distribution of individual savings, and it does not allow for every source of bias.

Sources: H8-A (the bar charts and distributions) §3 (the two-business example and the city summaries) and §4, Figure P (the pilot). Do not attach the landfill percentages or the pilot interval to the bar charts on page 1.